Engineering Mathematics-II (NMCI102),
Quiz-I, Max Marks 10
Winter Semester-2025-2026, IIT(ISM) Dhanbad
Date: 16.02.2026, Time: 30 min

Name: **Adm No:** **Section:**

A

Important notes / Exam rules

- Answer all the questions by choosing the correct answer. **There may be more than one correct answer for each question.**
- Each correct answer would fetch **one mark** and **NO** partial credit.
- Completely **Bubble the boxes** for the correct answer(s) only. **Overwriting/Half bubble/Other than bubbling such as tick mark** shall be awarded **ZERO** mark.
- Notations have their usual meaning.

Q 1 Which of the following is/are subspace(s) of the vector space $\mathbb{R}^3$?

■ $\{(x, y, z) \in \mathbb{R}^3 : x - y = 0\}$ ■ $\{(x, y, z) \in \mathbb{R}^3 : x + z = 0\}$
□ $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$ □ $\{(x, y, z) \in \mathbb{R}^3 : x + z = 1\}$

Q 2 Let $f : \mathbb{R}^2 \mapsto \mathbb{R}^3$ be a linear map defined as $f(x, y) = (x, x + y, y)$. Then the nullity of f is

□ 1 ■ 0 □ 2 □ 3

Q 3 Which of the following pair(s) of 2×2 matrices can NOT be obtained by applying elementary row operations on $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$?

□ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ □ $\begin{bmatrix} 1 & 0 \\ 0 & 16 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ □ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 8 \\ 0 & 1 \end{bmatrix}$ ■ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q 4 Suppose U and W are two dimensional subspaces of $\mathbb{R}^3$. Then the smallest possible dimension of $U \cap W$ is

□ 0 ■ 1 □ 2 □ 3

Q 5 Let $V = \{(x_1, x_2, x_3, \ldots x_{2026}) \in \mathbb{R}^{2026} : x_1 = x_2 = \cdots = x_{1013} = 0 \quad \text{and} \quad x_{1014} + x_{1015} \cdots + x_{2026} = 0\}$. Then the dimension of V is

□ 1014 □ 1013 □ 2026 ■ 1012

Q 6 The coordinate vector of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ with respect to the ordered basis of $\mathbb{R}^2$ consisting of vectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is

■ $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ □ $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ □ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ □ $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

Q 7 Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$. Then the dimension of column space of A is

□ 0 □ 1 ■ 2 □ 3

Q 8 Let $T : M_{2\times2}(\mathbb{R}) \to \mathbb{R}^2$ be a linear map defined by $T \begin{bmatrix} a & b \\ c & d \end{bmatrix} := \begin{bmatrix} a + d \\ b + c \end{bmatrix}, \quad \forall \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times2}(\mathbb{R})$. Then T is

□ neither one-to-one nor onto □ both one-to-one and onto
□ one-to-one but not onto ■ onto but not one-to-one

Q 9 Consider the following system of homogeneous linear equations

$$\begin{aligned} 6x_1 - x_2 + x_3 &= 0 \\ tx_1 + x_3 &= 0 \\ x_2 + tx_3 &= 0, \end{aligned}$$

where t is a parameter. Then which of the following statement is true for the system

- [ ] Unique solution for all values of t
- [ ] Non trivial solution for all values of t
- [ ] Non trivial solution only when $t = 0$
- [x] Non trivial solution only for $t = 2$ or $t = -3$

Q 10 Let $n \geq 3$ be an integer and $u_1, u_2, \ldots, u_n$ be n linearly independent elements in a vector space over $\mathbb{R}$. Set $u_0 = 0, u_{n+1} = u_1$ and define $v_i = u_i + u_{i+1}$ and $w_i = u_{i-1} + u_i$, for $i = 1, 2, \ldots, n$. Then

- [x] $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2025$
- [x] $\{w_1, w_2, \ldots, w_n\}$ is linearly independent if $n = 2025$
- [ ] $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2026$
- [x] $\{w_1, w_2, \ldots, w_n\}$ is linearly independent if $n = 2026$

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B

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Q 1 The coordinate vector of $\begin{bmatrix}2\\3\end{bmatrix}$ with respect to the ordered basis of $\mathbb{R}^2$ consisting of vectors $\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}3\\4\end{bmatrix}$ is

□ $\begin{bmatrix}2\\0\end{bmatrix}$ □ $\begin{bmatrix}1\\1\end{bmatrix}$ ■ $\begin{bmatrix}-1\\1\end{bmatrix}$ □ $\begin{bmatrix}3\\0\end{bmatrix}$

Q 2 Which of the following pair(s) of 2×2 matrices can NOT be obtained by applying elementary row operations on $I_2 = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$?

□ $\begin{bmatrix}1 & 0\\0 & 16\end{bmatrix}, \begin{bmatrix}1 & 0\\2 & 1\end{bmatrix}$ □ $\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}, \begin{bmatrix}1 & 8\\0 & 1\end{bmatrix}$ □ $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}2 & 0\\0 & 1\end{bmatrix}$ ■ $\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$

Q 3 Consider the matrix $A = \begin{bmatrix}1 & 2 & 3\\2 & 4 & 6\\1 & 1 & 1\end{bmatrix}$. Then the dimension of column space of A is

□ 0 □ 3 ■ 2 □ 1

Q 4 Let $f : \mathbb{R}^2 \mapsto \mathbb{R}^3$ be a linear map defined as $f(x, y) = (x, x + y, y)$. Then the nullity of f is

□ 1 □ 2 □ 3 ■ 0

Q 5 Let $T : M_{2\times2}(\mathbb{R}) \to \mathbb{R}^2$ be a linear map defined by $T\begin{bmatrix}a & b\\c & d\end{bmatrix} := \begin{bmatrix}a+d\\b+c\end{bmatrix}, \quad \forall \begin{bmatrix}a & b\\c & d\end{bmatrix} \in M_{2\times2}(\mathbb{R})$. Then T is

■ onto but not one-to-one □ one-to-one but not onto
□ neither one-to-one nor onto □ both one-to-one and onto

Q 6 Which of the following is/are subspace(s) of the vector space $\mathbb{R}^3$?

■ $\{(x, y, z) \in \mathbb{R}^3 : x - y = 0\}$ □ $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$
■ $\{(x, y, z) \in \mathbb{R}^3 : x + z = 0\}$ □ $\{(x, y, z) \in \mathbb{R}^3 : x + z = 1\}$

Q 7 Let $n \geq 3$ be an integer and $u_1, u_2, \ldots, u_n$ be n linearly independent elements in a vector space over $\mathbb{R}$. Set $u_0 = 0, u_{n+1} = u_1$ and define $v_i = u_i + u_{i+1}$ and $w_i = u_{i-1} + u_i$, for $i = 1, 2, \ldots, n$. Then

■ $\{w_1, w_2, \ldots, w_n\}$ is linearly independent if $n = 2026$ ■ $\{w_1, w_2, \ldots, w_n\}$ is linearly independent if $n = 2025$
■ $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2025$ □ $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2026$

Q 8 Consider the following system of homogeneous linear equations

$$\begin{aligned} 6x_1 - x_2 + x_3 &= 0 \\ tx_1 + x_3 &= 0 \\ x_2 + tx_3 &= 0, \end{aligned}$$

where t is a parameter. Then which of the following statement is true for the system

- [x] Non trivial solution only for $t = 2$ or $t = -3$
- [ ] Non trivial solution for all values of t
- [ ] Unique solution for all values of t
- [ ] Non trivial solution only when $t = 0$

Q 9 Let $V = \{(x_1, x_2, x_3, \ldots x_{2026}) \in \mathbb{R}^{2026} : x_1 = x_2 = \cdots = x_{1013} = 0 \quad \text{and} \quad x_{1014} + x_{1015} \cdots + x_{2026} = 0\}$. Then the dimension of V is

- [x] 1012
- [ ] 1013
- [ ] 1014
- [ ] 2026

Q 10 Suppose U and W are two dimensional subspaces of $\mathbb{R}^3$. Then the smallest possible dimension of $U \cap W$ is

- [ ] 3
- [x] 1
- [ ] 2
- [ ] 0

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■ $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2025$
■ $\{w_1, w_2, \ldots, w_n\}$ is linearly independent if $n = 2025$
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■ $\{(x, y, z) \in \mathbb{R}^3 : x + z = 0\}$

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■ $\{w_1, w_2, \ldots, w_n\}$ is linearly independent if $n = 2025$

■ $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2025$

□ $\{v_1, v_2, \ldots, v_n\}$ is linearly independent if $n = 2026$

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□ 1014 □ 1013 ■ 1012 □ 2026

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